



Interior Display System Proposal

Powered by CoachMate

Prepared for: **[Employer Name]**

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Confidential — Prepared for Internal Review

1. Executive Summary

This proposal, powered by CoachMate, introduces a modern interior passenger information display system designed specifically for older fleet vehicles such as the Volvo B7RLE. The system uses a controller-driven HDMI display, bypassing legacy DG3/IBIS hardware entirely. This delivers a passenger information experience using UK-sourced components, with minimal electrical integration and full control over the UI.

The solution is:

- **Modern** — full graphics, animations, branding
- **Compatible** — works with any older bus
- **Maintainable** — uses standard, UK-stocked components
- **Cost-effective** — avoids EG4/Ethernet retrofits
- **Deterministic** — the Bus Controller drives all display logic
- **Regulation-aware** — designed against the PSV Accessible Information Regulations 2023 display requirements; see Appendix A for the full compliance mapping

2. System Architecture Overview

High-Level Architecture

```
24V Bus Supply --> Victron Orion-Tr Isolated 24/12 DC-DC Converter --> Fused 12V  
Distribution --> [Bus Controller, Interior Display, Driver PWA charging] --HDMI--> Interior  
Display
```

Application Flow

Operations PWA Creates routes, duties, schedules. Sends duty link to driver.

Driver PWA Receives duty link. If none: shows Route Selector. On Start: begins GPS polling and progress tracking. Joins the Bus Controller's local WiFi access point and sends structured progress events to it over that link.

Bus Controller Hosts a local WiFi access point for the Driver PWA to join. Receives progress events. Renders final UI. Outputs HDMI to interior display.

Interior Display Shows the passenger information UI.

3. Driver PWA (Dedicated Chapter)

The heart of the platform's real-time intelligence

The Driver PWA is the operational core of the ecosystem. It is responsible for:

- Duty retrieval
- Route selection
- GPS polling
- Stop detection
- Progress tracking
- Sending structured progress events to the Bus Controller

It runs on a dedicated tablet mounted in the driver's cab.

3.1 Driver PWA Responsibilities

Duty Handling

- Receives duty link from Operations PWA
- Displays duty summary
- If no duty: presents Route Selector
- Ensures driver starts correct route

GPS Polling

- Polls GPS at fixed intervals
- Calculates distance to next stop
- Determines stop arrival/leave events
- Detects terminus arrival

Progress Event Generation

Driver PWA sends structured JSON events to the Bus Controller, including:

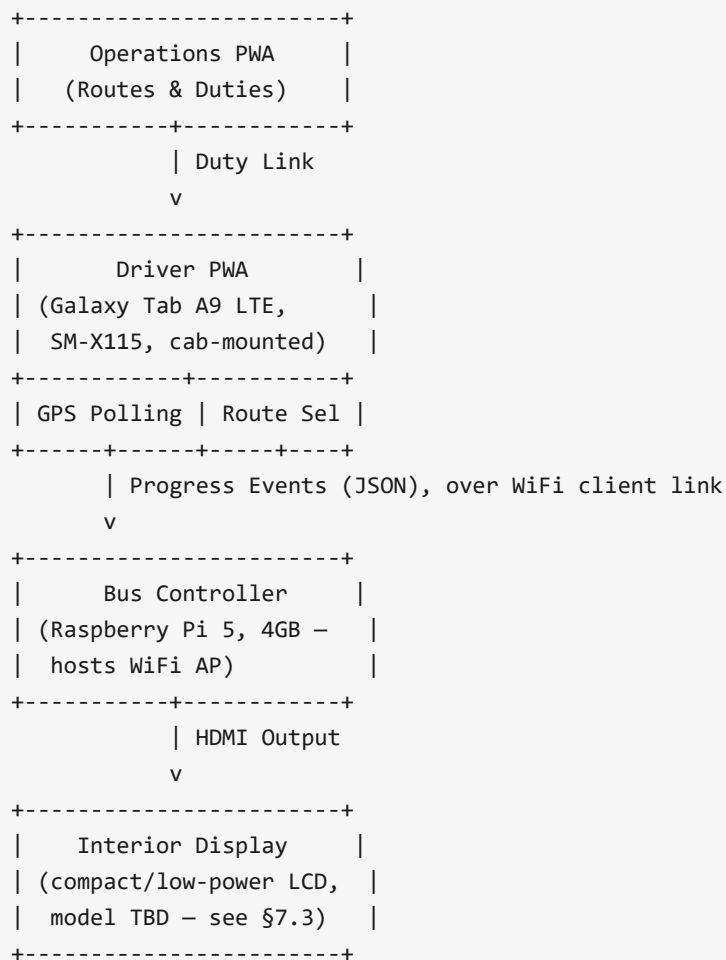
- `next_stop`
- `following_stop`
- `distance_to_next`
- `eta_next_stop`
- `progress_percent`
- `stop_requested`
- `disruption_message`
- `terminus_flag`

Connectivity

Corrected: the Bus Controller hosts the local network (a hostapd WiFi access point); the Driver PWA device joins it as a standard, low-privilege WiFi client. The device does not host its own hotspot.

- Joins the Bus Controller's WiFi access point as a client
- No pairing step required on the tablet beyond the one-time WiFi join
- USB tether (optional, bench/diagnostic use only)

3.2 Driver PWA Data Flow Diagram



4. Refined UI Mockups (Pixel-Level Specification)

Open item: a large-format 28" stretched-bar panel was evaluated but is not being pursued — sourcing it within the project timeline proved impractical, and more fundamentally, the target fleet (10+ year old vehicles, e.g. Volvo B7RLE) has interior wiring that cannot support a large-format TFT retrofit without a major rewiring job. The plan is now a smaller, lower-power "dumb" panel — still fully driven by the Bus Controller over HDMI, with the same custom UI — that fits within the existing wiring's practical limits. Panel model is not yet chosen (see §7.3). The layout below is vh-based, so it is written to scale to whatever panel is finally selected, but the specific pixel/physical dimensions below (assumed for a 28"/16:3 panel) will need re-checking against the real chosen panel before build.

4.1 Layout Zones

Zone A — Header Bar (Top 12%)

- Background: #0D0D0D
- Route number: Roboto Bold, 72px, #F9A825
- Destination: Roboto Medium, 48px, #FFFFFF
- Time: Roboto Regular, 36px, #CCCCCC
- Operator logo: white silhouette
- Company logo placed top-left in header

Zone B — Main Content (Middle 65%)

Next Stop (Primary)

- Font: Roboto Black, 96px, #FFFFFF
- Alignment: centred
- Animation: slide-in (300ms)

Following Stop (Secondary)

- Font: Roboto Regular, 48px, #AAAAAA
- Alignment: centred
- Animation: fade-in (200ms)

Icon Row

- Icons: wheelchair, bell, interchange
- Style: line icons, 2px stroke
- Colour: #F9A825
- Size: 48px

Zone C — Footer Bar (Bottom 23%)

Progress Bar

- Background: #1A1A1A
- Bar colour: #F9A825
- Height: 24px
- Animation: smooth linear progression

Disruption / Notices

- Font: Roboto Medium, 36px, #FF5252
- Behaviour: static or scrolling

4.2 Colour Palette

Purpose	Colour
Background	#0D0D0D
Primary text	#FFFFFF
Secondary text	#AAAAAA
Accent	#F9A825
Warning	#FF5252

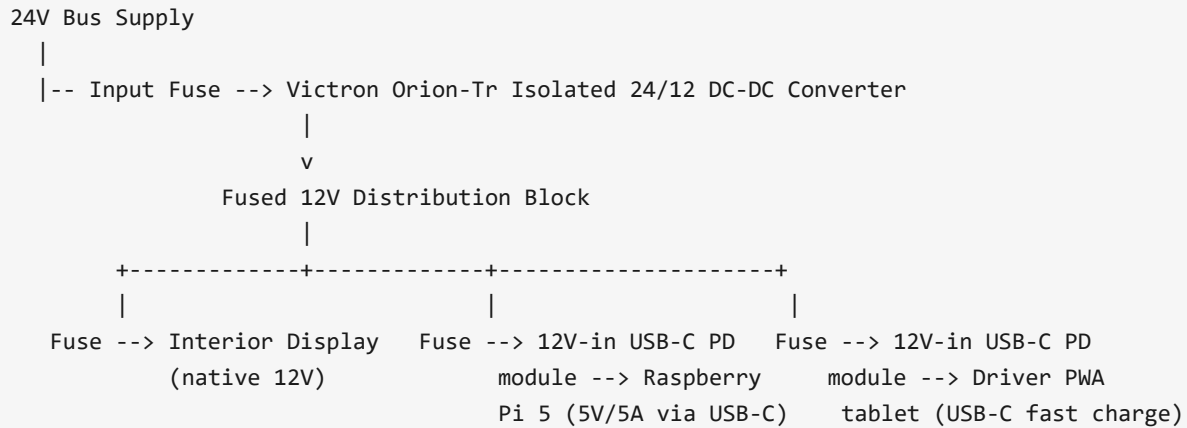
4.3 Typography

- Roboto family (Bold, Medium, Regular, Black)
- High readability
- Clean, modern, widely supported
- Works well on standard LCD panels

5. Wiring Diagram (Text-Based)

Corrected: power conditioning is a single isolated 24V→12V converter stage feeding a fused 12V distribution block, with point-of-load USB-C PD modules per device — not separate 24V→12V and 24V→5V converters.

Power Path



Signal Path

```
Driver PWA --> Progress Events (JSON, over WiFi client link) --> Bus Controller
Bus Controller HDMI Out --> Shielded HDMI Cable --> Interior Display (native HDMI input)
```

Grounding

```
Interior Display GND
Bus Controller (Pi) GND
DC-DC Converter GND
--> Single Chassis Ground Point
```

6. Installation Plan (Volvo B7RLE)

6.1 Mechanical Installation

1. Identify roof structure/cross-members above the aisle, just past the entry area.
2. Install a ceiling-drop mount sized to the final panel's weight and VESA/mounting pattern (TBD — see §7.3) with bolt-through fixings into the roof structure.
3. Add anti-vibration isolation mounts.
4. Mount the Interior Display to the ceiling-drop mount.
5. Install the Bus Controller (Raspberry Pi 5) and power electronics together in a single **polycarbonate** enclosure, recessed into the same ceiling void. Polycarbonate is specifically required, not metal — a metal enclosure would attenuate the Pi's onboard WiFi (Faraday-cage effect), which the Driver PWA depends on to reach the access point it hosts.
6. Install the Driver PWA tablet mount in the cab using a RAM Mounts C-size ball-and-socket cradle on a bolted (not suction-cup) base, rated for commercial-vehicle vibration.

6.2 Electrical Installation

1. Tap into auxiliary 24V circuit, downstream of the vehicle's master isolator switch.
2. Install input fuse ahead of the Victron Orion-Tr Isolated 24/12 converter, and an output fuse per branch off the 12V distribution block.
3. Mount the converter and fused distribution block in the enclosure.
4. Run 12V to the Interior Display (native 12V input).
5. Run 12V to a USB-C PD module powering the Raspberry Pi 5 (5V/5A over USB-C).
6. Run 12V to a USB-C PD module for Driver PWA tablet charging at the cab mount.
7. Install ferrite chokes on power lines.
8. Run shielded HDMI cable from the Bus Controller to the Interior Display.

6.3 Software Commissioning

1. Install Raspberry Pi OS, boot from USB3/NVMe SSD (avoids SD-card corruption risk on ungraceful power loss).
2. Configure hostapd WiFi access point on the Pi (with the WiFi interface marked `unmanaged` in NetworkManager so it doesn't fight the manually-run `hostapd`), and configure auto-boot systemd services for both the AP and the kiosk browser.
3. Pair the Driver PWA tablet to the Pi's access point (one-time WiFi join at commissioning).
4. Set HDMI output to the chosen panel's native resolution.
5. Test UI transitions and brightness for daylight visibility through the vehicle glazing.
6. Perform road test for readability, stability, and AP reconnect behaviour after a power cycle.

7. Procurement Tables (All Devices)

7.1 Driver PWA Device

Corrected: the confirmed device is a tablet, not a phone — specifically the LTE SKU, which is the only variant with a GNSS chip.

Item	Model	Supplier	Notes
Driver PWA Device	Samsung Galaxy Tab A9, LTE (SM-X115), 8GB/128GB	Samsung UK / Currys	LTE SKU required — the Wi-Fi-only SM-X110 has no GNSS chip and would fail the geofence requirement. LTE also covers mid-route Supabase connectivity independent of the Pi's local AP.
Cab Mount	RAM Mounts C-size ball-and-socket, X-Grip or Tab-Tite cradle	RAM Mounts UK (ram-mount.co.uk) or MUD-UK	Bolted base, not suction-cup — rated for sustained commercial-vehicle vibration

7.2 Bus Controller

Item	Model	Supplier	Notes
Bus Controller	Raspberry Pi 5, 4GB	The Pi Hut	4GB is sufficient for this workload (kiosk browser + hostapd + one WebSocket link); HDMI output
Storage	USB3/NVMe SSD (boot device)	The Pi Hut	Preferred over SD card — avoids the classic SD-corruption failure mode on ungraceful power loss, using Pi 5's PCIe support
Enclosure	Polycarbonate, combined with power electronics	—	Not metal — metal would attenuate the Pi's onboard WiFi needed to host the access point

7.3 Interior Display

Open item — panel not yet selected. A 28" stretched-bar panel (Allsee WS28HD8-B) was the earlier candidate but is no longer being pursued: it proved difficult to source within the project timeline, and — more fundamentally — the target fleet (10+ year old vehicles) has interior wiring that cannot support a large-format TFT retrofit without a major rewiring job. The direction now is a smaller, lower-power, easier-to-source "dumb" panel that still plugs into the same Bus Controller over HDMI and renders the same custom UI, but fits within what the existing wiring can support without a full rewire. Candidate panels are being evaluated; model, exact power draw, and mounting pattern are TBD.

Item	Model	Supplier	Notes
Display Panel	TBD — compact HDMI-input LCD, low power draw, standard aspect ratio	Being sourced	Selection criteria: HDMI input, power draw compatible with the vehicle's existing sign wiring capacity (no major rewire), physically small enough for the ceiling-void

			install. Once chosen, re-verify against Appendix A §1.1–1.4 (seat visibility, text height).
Ceiling Mount	TBD — sized to final panel's weight/mounting pattern	—	To be selected once the panel is confirmed

7.4 Power System

Corrected: a single isolated converter stage plus point-of-load modules, replacing the two separate generic 24V→12V/24V→5V converters.

Item	Model	Supplier	Notes
Central DC-DC Converter	Victron Orion-Tr Isolated 24/12	12 Volt Planet / Sunshine Solar / BMS Technologies / Midsummer Energy (UK)	Isolated variant specifically — galvanic isolation gives real noise/interference blocking, sold explicitly for comms/entertainment/navigation systems. Feeds one clean fused 12V distribution point.
Point-of-load Modules	12V-input USB-C PD modules, 25–60W rated	RS / Farnell	One per device (Pi 5, Driver PWA tablet); Interior Display takes 12V directly, no conversion needed
Fuses	Sized per branch off the distribution block	Vehicle Wiring Products	Individual fuse per branch — one downstream short doesn't take out the rest

7.5 Cabling & Mounting

Item	Model	Supplier	Notes
HDMI Cable	Shielded, short run	The Pi Hut	Reduce interference
Automotive Wiring	24V rated	Vehicle Wiring Products	Power loom
Mounting Hardware	M8 bolts, anti-vibration isolation mounts	Screwfix	Roof/ceiling fixing

8. Project Timeline

Phase	Duration
Panel selection (evaluating alternatives to the earlier 28" candidate)	TBD
Procurement (once panel confirmed)	1-2 weeks
Installation (per bus)	3-5 hours
Software commissioning	1 hour
Road testing	1 day

Timeline depends on finalising the panel choice first — the earlier 28" candidate's B2B lead time is no longer the constraint, but a replacement panel meeting the wiring/power constraints hasn't been confirmed yet.

9. Cost Summary

Costs to be confirmed once a panel is selected and quoted, along with the Victron converter and mounting hardware — no placeholder figures are given here to avoid understating or overstating cost ahead of that decision.

10. Benefits to the Operator

- Modern passenger experience
- Full control over UI and branding
- No dependency on DG3/IBIS
- No expensive EG4/Ethernet retrofit
- Uses standard UK-available/UK-stocked components throughout
- Easy maintenance and replacement
- Scalable across entire fleet
- Designed against the PSV(A)R 2023 display requirements — see Appendix A

11. Conclusion

This system provides a modern, reliable, and cost-effective interior display for older buses. It leverages your existing software, UK-sourced components confirmed against real supplier availability, and a clean HDMI architecture to deliver a passenger experience without the complexity of legacy bus systems.

Appendix A: PSV Accessible Information Regulations (PSV(AI)R) Compliance

Prepared against two source documents supplied by the operator: the CPT Operator Checklist for Vehicle Compliance (display, audio, and announcement requirements), and the accompanying CPT guidance note on diversions, hail-and-ride, and Annexes A (volume measurement) and B (alerts).

This appendix covers the **full** PSV(AI)R scope. The Interior Display System described in the body of this proposal delivers the **visual display** rows (Section A1) and receives the disruption/diversion text shown on-screen. Audio announcements, volume calibration, and driver-triggered diversion alerts are delivered by the separate Driver PWA announcement subsystem (pre-rendered TTS clips + live speech synthesis), not by the Interior Display hardware covered above — rows below are marked accordingly.

Status key: **Compliant** · **Partial** · **Gap** · **Planned** · Out of scope (hardware/procedural) · N/A

A1. Display requirements (visual) — delivered by this proposal's Interior Display system

#	Requirement	Status	Notes
1.1	Visible from ≥51% of passenger seats on each deck (no standing passengers)	Planned	Central-aisle ceiling mounting position is the intended approach; the panel itself is not yet chosen (see §7.3), so seat-visibility percentage cannot be confirmed until a candidate panel's size/position is finalised.
1.2	Visible from each forward-facing wheelchair space, and any designated priority seats	Planned	Same as 1.1 — depends on the final panel choice and position, not yet confirmed.
1.3	Buses first used from 1/10/24: display reasonably visible from rearward-facing wheelchair space(s)	Planned	Not yet verified — depends on final panel position relative to each vehicle's wheelchair bay, which varies by vehicle model.
1.4	Display text ≥22mm in height, on a contrasting background	Planned	The UI layout (Section 4) is height-relative (vh-based) and the colour palette (Section 4.2) is high-contrast, so the design intent supports this requirement on any reasonably-sized panel. However, the earlier "compliant by design" reasoning specifically relied on the 28" panel's fixed physical height — that panel is no longer the plan (see §7.3), so 22mm compliance must be re-verified once the replacement panel's physical size is confirmed. Do not treat as settled.
1.5	Audio announcements must be consistent with	Gap	The Interior Display currently renders from its own progress-event feed; the live link from the Driver PWA's announcement engine to

	the visual information provided		the Interior Display so the two are guaranteed to show/say the same thing at the same moment is not yet built/verified end-to-end.
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A2. Audio requirements — delivered by the Driver PWA announcement subsystem, not this proposal's hardware

#	Requirement	Status	Notes
2.1	Pre-measured ambient / live-measured volume levels measured per Annex A	Out of scope (procedural)	Physical calibration exercise (empty vehicle, front/rear average, 5mph & 20mph) carried out by ops staff with a sound-level meter. The Driver PWA platform stores the resulting figure (vehicle_audio_config: ambient reading, fixed output level, measured-by, measured-at) but does not perform the measurement. Needs a named owner and documented procedure.
2.2	Audio delivered at every seat/wheelchair space at a level 3dB louder than ambient (non-adaptive) or live ambient (adaptive)	Out of scope (hardware)	Function of the vehicle PA amplifier/speaker installation and its physical calibration — outside this proposal's scope (this proposal covers the visual display only). No PA/amp/speaker system is currently specified.
2.3	Audio not delivered louder than 84dB at any seat/wheelchair space	Out of scope (hardware)	Same as 2.2 — a physical PA/amplifier ceiling, needs a named hardware owner.
2.4	Volume control facility — adaptive (auto-varies with live ambient) OR non-adaptive (pre-measured, operator-set)	Compliant — non-adaptive route	The Driver PWA platform implements the non-adaptive path: one fixed calibrated output level per vehicle, applied uniformly. Valid under the regulation's non-adaptive option.
2.5	Audio available via induction hearing loop at every wheelchair space and priority seat, for telecoil hearing aids	Out of scope (hardware)	Vehicle audio-hardware component, independent of this proposal and the Driver PWA software. Not addressed anywhere in the current build. Needs a named hardware owner and confirmation of fitment per vehicle.
2.6	Operators must not require use of a "personal electronic device" to receive the mandated information	Compliant	All mandated audio/visual info is delivered via vehicle-mounted hardware (this proposal's Interior Display, and the Driver PWA driving the PA). No passenger-facing app or personal device is required.

A3. Announcement content & timing requirements — delivered by the Driver PWA announcement subsystem

#	Requirement	Status	Notes
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3.1	At each stopping place, beginning when doors open: announce route name/number and final stopping place (or direction)	Compliant	Implemented, announcing route code and final destination on arrival. Trigger nuance: fires on GPS-confirmed arrival, not a physical door-open sensor — there is no door-sensor integration in the current hardware stack, so "doors open" and "GPS arrival" are treated as equivalent triggers.
3.2	After leaving/passing each stopping place, in good time before the next: announce the name of the next stopping place	Compliant	An "approaching" announcement fires 250m out from the next stop, ahead of arrival.
3.3	Passengers informed when the service has reached its final stopping place, preceded by an alert	Partial	The final-stop announcement itself is implemented and fires once, on arrival. No distinct "alert" sound precedes it — see A4, row 5.1.
3.4	Sufficient time provided at the final stopping place for passengers to alight	Compliant	The final-stop announcement is not time-boxed or auto-dismissed; nothing in the software curtails alighting time.
3.5	Facility to provide diversion information to passengers	Compliant	Driver-triggered "Diversion Alert" control in the Driver PWA fires an immediate fixed announcement and suppresses normal stop announcements while active.
3.6	Diversion known in advance: information provided before the last scheduled stopping place, prior to the diversion commencing	Gap	Only real-time, driver-toggled alerts exist — there is no concept of a pre-planned diversion with route/timing data the system could use to announce automatically ahead of the diversion point. Recommend scoping this as a follow-up, or formally confirming driver procedure as the interim mitigation.
3.7	Diversion not known in advance: information given as soon as possible once known	Compliant	Exactly what the driver-triggered button provides.
3.8	Diversion info preceded by an alert	Gap	Same alert-tone gap as 3.3/5.1.
3.9–3.12	Hail-and-ride notification requirements (start of route, mid-route commencement, end-of-section warning, preceded by an alert)	N/A	Confirmed with the operator: no current or planned routes use hail-and-ride sections. If this changes, none of the schema, GPS engine, or announcement pipeline currently models a hail-and-ride section — it would need real design/build work.

A4. Annex A (volume measurement) & Annex B (alerts)

#	Requirement	Status	Notes
4.1	Annex A — non-adaptive/adaptive	Out of scope (procedural)	See A2, row 2.1 — operational process, not software.

	measurement methodology (empty vehicle, front/rear average, 5mph & 20mph)		
5.1	Annex B — where an alert is required, it is "a sound intended to draw the attention of passengers to the information following it"	Gap	No distinct alert tone/chime exists anywhere in the current audio pipeline. Diversion, final-stop, and (if applicable in future) hail-and-ride announcements go straight into spoken content with nothing preceding them. Recommend scoping a short fixed chime clip ahead of these three announcement types — ordinary next-stop/approaching announcements are not "alerts" under the regulation and don't need one.

Summary of open items

Real gaps requiring build work:

- **5.1 / 3.3 / 3.8** — no alert tone precedes diversion or final-stop announcements.
- **3.6** — no system support for advance-notice diversions (only real-time driver-triggered).
- **1.5** — visual/audio consistency can't be verified until the Interior Display is confirmed receiving live, synchronised state alongside the Driver PWA's spoken announcements.

Hardware/installation work, not software gaps — needs a named owner and sign-off:

- Section A1 (display) — the interior display panel itself is not yet chosen (see §7.3): the earlier 28" candidate was dropped due to sourcing difficulty and the target fleet's wiring being unable to support a large-format TFT retrofit without a major rewire. 1.1–1.4 all depend on this decision and must be re-verified once a replacement panel is confirmed.
- 2.2 / 2.3 / 2.5 — PA amplifier calibration (3dB above ambient, 84dB ceiling) and induction hearing loop fitment; no PA/amp system is currently specified in this proposal.
- 2.1 / 4.1 — the physical ambient-measurement procedure itself.

Confirmed not applicable: hail-and-ride (3.9–3.12) — no routes use hail-and-ride.

Compliant / acceptable as designed: 2.4 (non-adaptive volume control), 2.6 (no personal device required), 3.1, 3.2, 3.4, 3.5, 3.7.